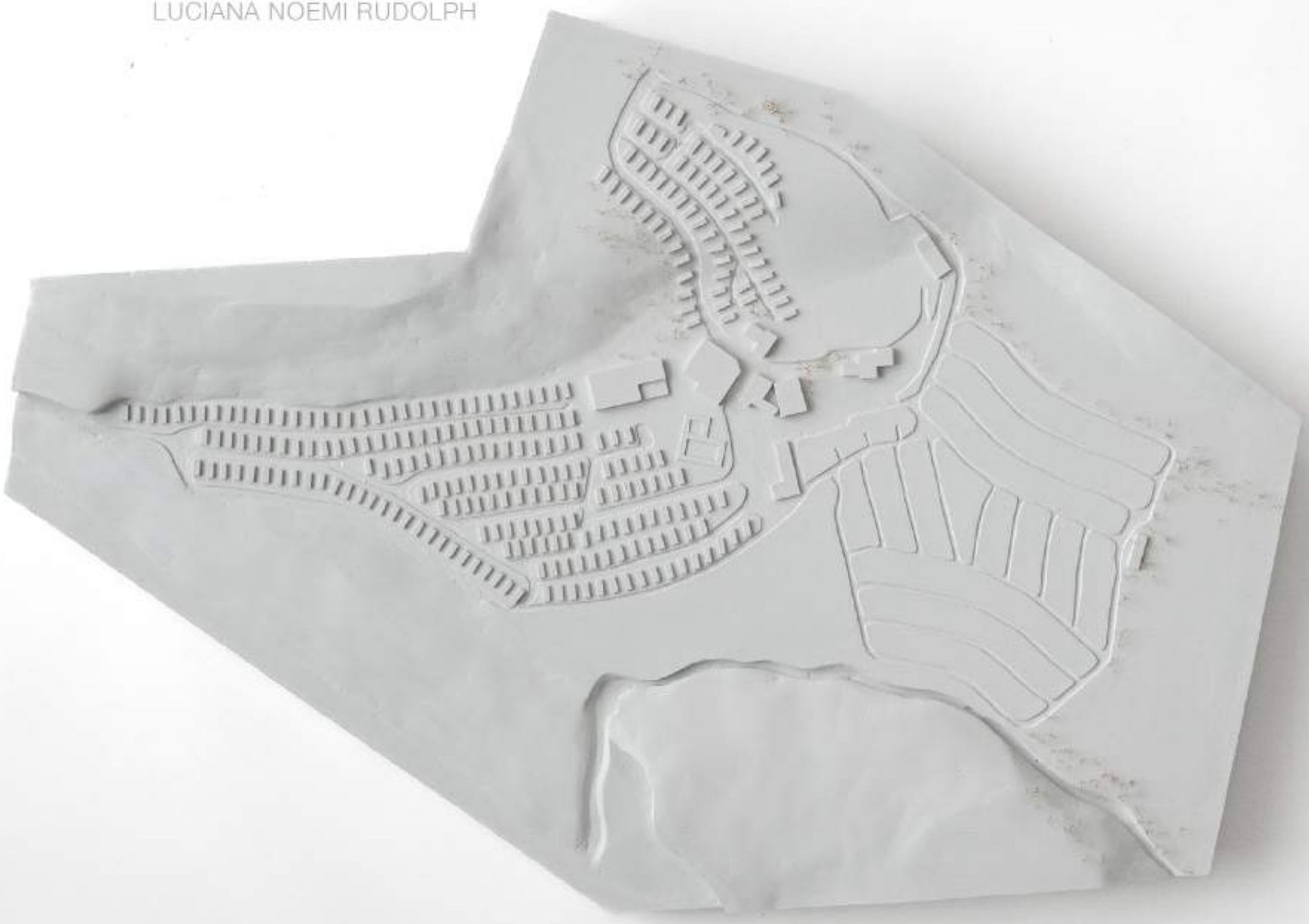
PORTFOLIO

LUCIANA NOEMI RUDOLPH



01 IPAI COMMUNICATION CENTER
Design Assistant at MVRDV

02 RECYCLE RESIDENCE
Prof. Stephan Birk, Independent Bachelor's Thesis, 8th Semester

03 INSIDE OUT
Visiting Studio Schneider Türtscher, 7th Semester

04 URBAN WETLANDS
Studio Krucker Bates, 4th Semester

05 THE TETRIS HOUSE
Prof. Florian Musso 3rd Semester

Luciana Noemi Rudolph | +49 157 38206686 | Schraudolphstrasse 12 | 80799 Munich

IPAI COMMUNICATION CENTER

01

Design Assistant at MVRDV, DAS Studio
Shandelys Job, +31104773860

The Communication Centre is a key component of MVRDV's visionary IPAI project in Heilbronn. As a Design Assistant at MVRDV within the DAS Studio in Rotterdam, I contribute specifically to this central hub of the AI campus, which serves as the core platform for information exchange and public engagement.

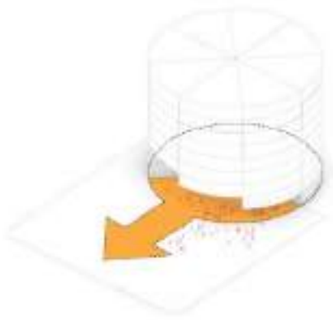
Working within a dynamic, multidisciplinary team through performance phases 2 and 3, I focus on the conceptual development and architectural refinement of the Communication Centre. My responsibilities include producing high-quality renderings and diagrammatic visualizations that effectively communicate our design intentions to the clients – IPAI and Schwarz Group. Currently, our team is refining two distinct facade concepts that respond to the building's cylindrical form while integrating AI generated LED elements that interact dynamically with the urban environment. I create comprehensive interior and exterior renderings of these facade alternatives, ensuring that our proposals convey both technical sophistication and a compelling public narrative in client presentations.

Envisioned as a dynamic gateway, the Communication Centre redefines the flow of public movement around the building. Its design incorporates cutting-edge interactive technologies to establish an engaging and sustainable urban landmark. Our work exemplifies MVRDV's commitment to socially and environmentally responsible architecture, positioning the Communication Centre as an innovative and contextually integrated element of the IPAI Campus project.

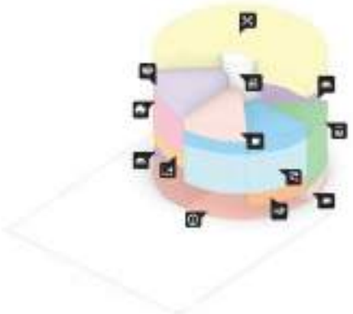
MY PARTICIPATION IN THE PROJECT:

Assisting in facade design, conceptual diagrams, interior and exterior rendering with Rhino+Revit models in Enscape, isometric drawings and facade sections in Rhino+illustrator, 3D facade modeling in Grasshopper.

1. CYLINDRICAL BASE FORM

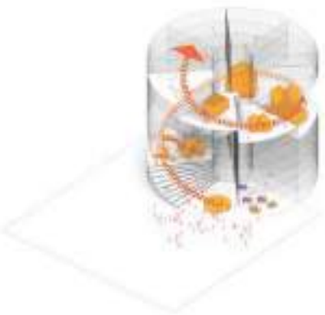


2. CONNECTION TO PUBLIC SPACE

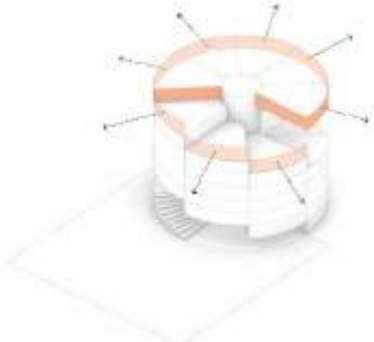


3. STACKED PROGRAM

SITE PLAN 1:500, MADE BY MVRDV COLLEGUES

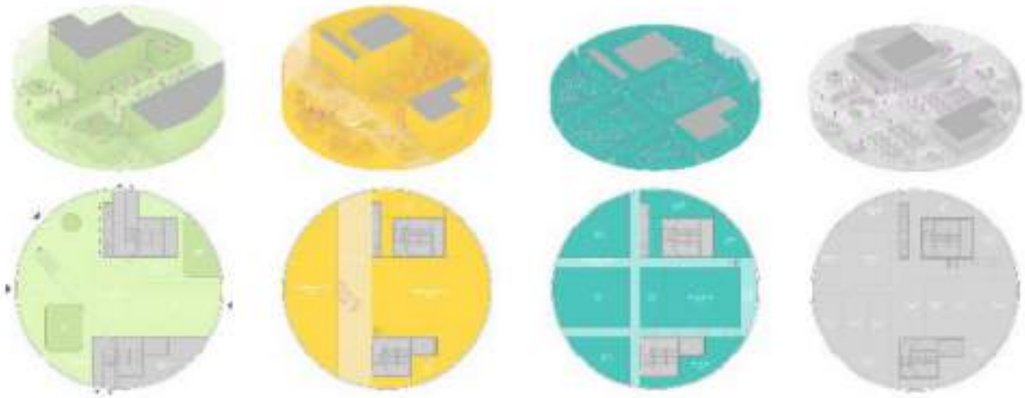


4. SPIRAL LIVING ROOM



5. VIEWING PLATFORMS

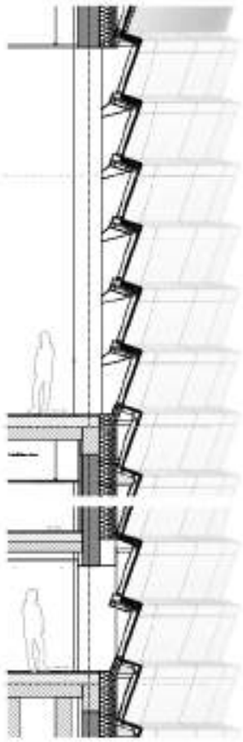
6. INTERACTIVE FACADE



FLOOR PLANS 1:200
DAY AND NIGHT RENDER MADE BY LUCIANA R.
(RHINO+GRASSHOPPER+ENSCAPE)

SECTION 1:100, MADE BY MVRDV COLLEGUES

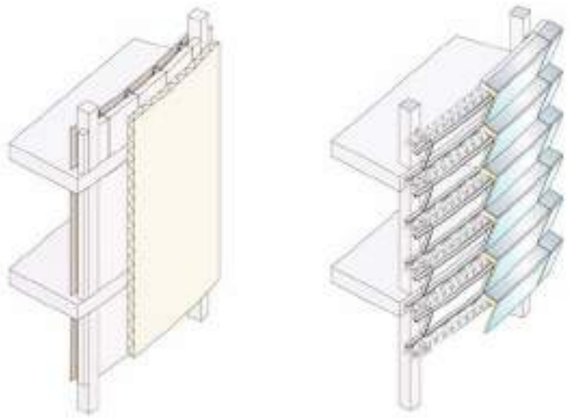
The spiraling 'living room' forms the heart of the Communication Center, seamlessly connecting conference rooms, co-working spaces, and exhibition areas while adapting in height to suit each function. This dynamic spatial flow fosters interaction and innovation, embodying the building's role as a hub for AI-driven research and collaboration. The facade further enhances this identity, integrating responsive elements that reflect the ever-evolving nature of artificial intelligence.



SECTION FACADE DETAILS 1:25
BY LUCIANA AND ALEX N.

FIRST FACADE OPTION

There are currently two facade options under consideration, each evaluated based on key factors such as monolithic aesthetics, cost efficiency, sustainability, and maintenance requirements. The first option, inspired by the Depot Boijmans Van Beuningen, features mirrored glass panels that reflect the surrounding environment while gradually transitioning to a more transparent finish around window openings to allow visibility.

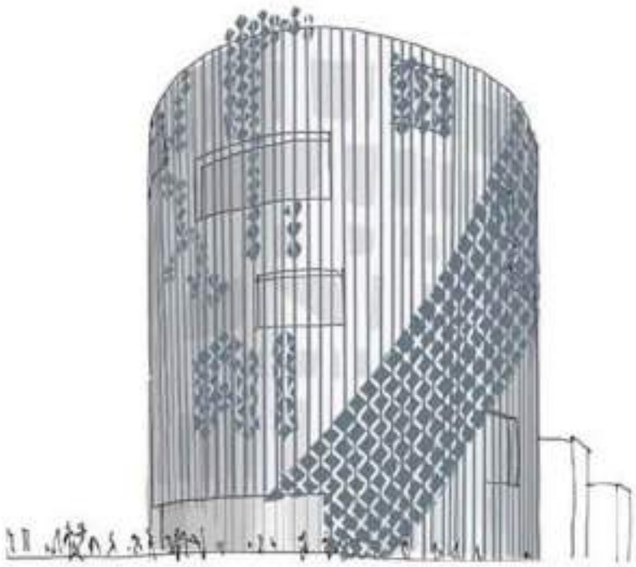


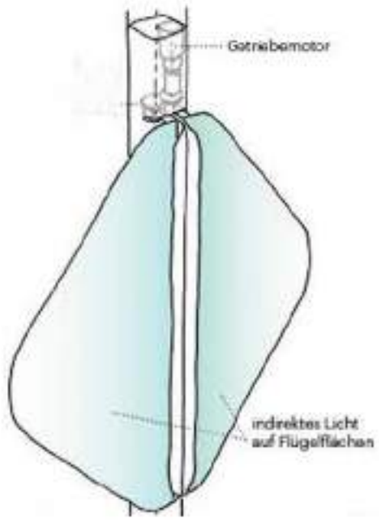
3D RHINO MODEL PARTIALLY MODELED BY LUCIANA R., RENDER FROM VIVID

INTERIOR RENDER BY LUCIANA R. (RHINO+ENSCAPE)

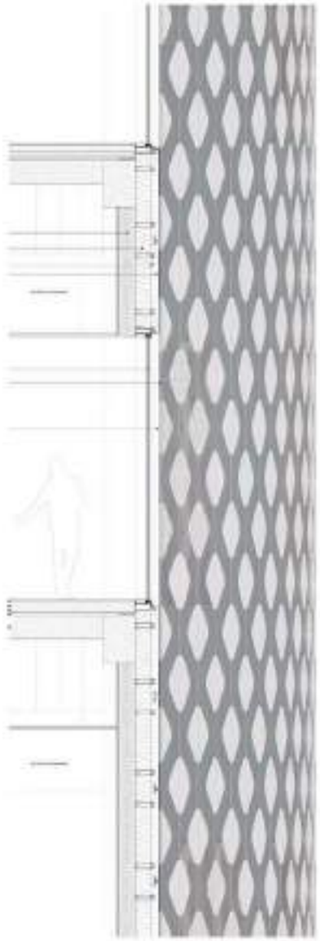
SECOND FACADE OPTION

The second facade option combines sustainability with AI-driven adaptability, featuring aluminum panels with integrated photovoltaics and LEDs. These panels generate energy while displaying AI-generated content, creating a dynamic interplay of light and information that reinforces the Communication Center's innovative identity. Optionally, these „butterfly" panels can dynamically open and close for sun shading.

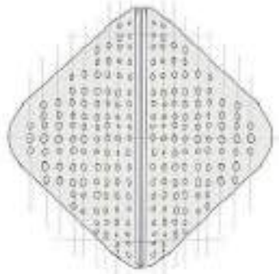




BUTTERFLY CONCEPT SKETCHES
INTERIOR RENDER BY LUCIANA R.



FACADE MEETS ROOFTOP RENDER
SECTION FACADE DETAILS 1:25
BY LUCIANA AND ALEX N.



PARTIALLY PERFORATED BUTTERFLY
WITH INTEGRATED PV AND LED STRIPE

REFLECTIVE BUTTERFLY MATERIAL
OPTIONAL: OPAQUE OR TRANSPARENT

RECYCLE RESIDENCE

02

Prof. Stephan Burk, +49 8928925491
Independent Bachelor's Thesis,
8th Semester

The "Recycle Residence" project is a practical design proposal for a two-story single-family home constructed entirely from repurposed materials, embracing the principle of "Form follows availability." As part of my individual bachelor's thesis, I developed a sustainable housing solution tailored for a real family, utilizing only materials already in circulation. The project centers on the innovative and responsible use of existing resources.

Key materials include "Home Bricks" made from donated wood scraps from prefabricated homes, as well as mahogany parquet flooring salvaged from the deconstruction of the former church in Füssen. The design features a two-story, raised structure with two distinct building sections, organized with a clear floor plan that highlights the materials themselves. The project was developed in close collaboration with specialized companies to ensure smooth implementation on-site.

This project is being realized in partnership with Supertecture, a non-profit organization focused on social and sustainable architecture, and is located in Irsee, Bavaria. The house is designed for a family of four facing unique health challenges. Through volunteer-based initiatives, Supertecture promotes creative building techniques and material reuse, working toward long-term solutions to ecological matters.

THE PROJECT WAS INDIVIDUAL WORK.

3D model and floor plans in ArchiCAD and Rhino, conceptual design, physical modelmaking, photography and editing by me, business development together with Till Lill from Supertecture.

Resources are used intentionally and sparingly, while the floor plan remains clear and well-organized. This deliberate reduction follows the principle of "Less is more." The intentional simplicity of the forms shifts the focus to the materials themselves, emphasizing their inherent value and aesthetic qualities.

The design emphasizes clean, simplified forms while avoiding unnecessary complexity in construction. This approach reduces material and labor requirements, enabling the participation of volunteer helpers. "Building Simply" represents a resource-efficient architecture that stands out for its accessibility and clarity.

Material choices in the design are closely tied to their availability within the circular economy. The design adapts flexibly to the properties of the materials, thoughtfully integrating the unique characteristics of reused elements into both the design and the construction process.

THE THREE DESIGN PRINCIPLES
ENGRAVED ON RECYCLED WOOD OFFCUTS

1 INDUSTRIAL SURPLUS

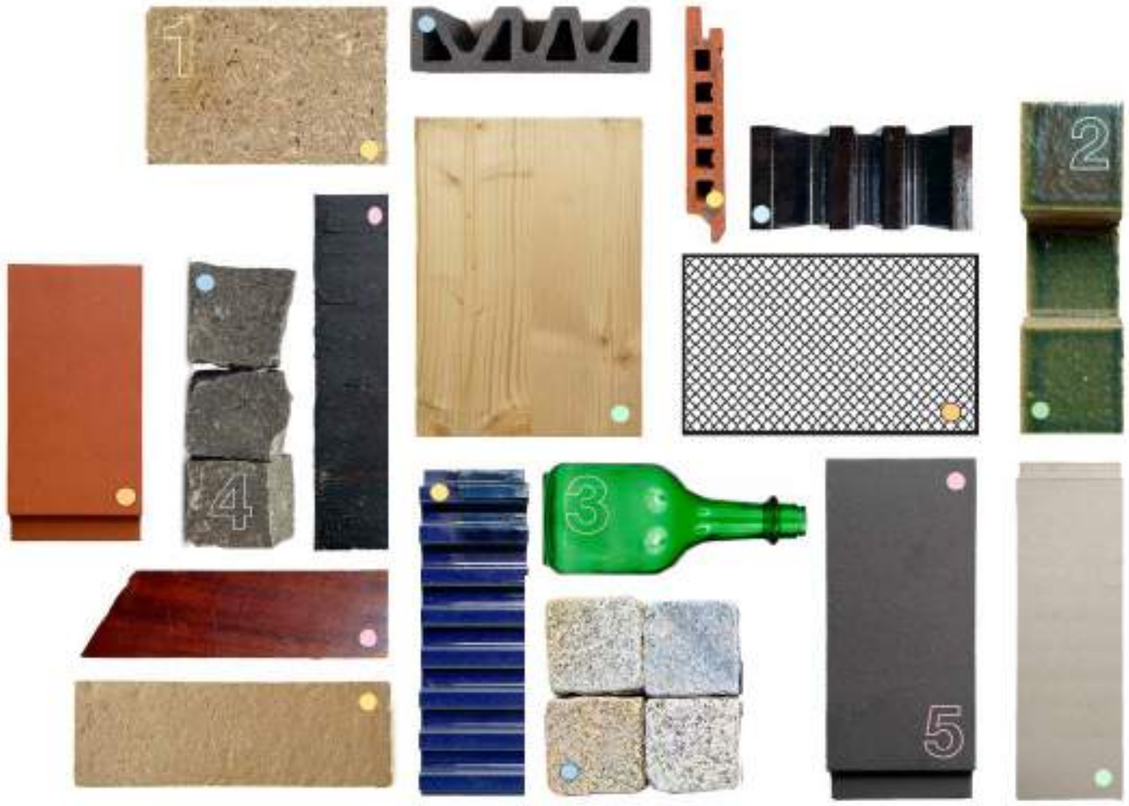
- Overproduction
- Recejted products/ second-choice material
- Non-construction material (e.g. walls from bottles)

2 TEMPORARY STRUCTURES

- Exhibition pieces
- Trade fair construction
- Temporary cultural structures (e.g. theater stage)

3 UPCYCLING

- Repurposed material from renovation projects
- Waste collection sites
- Non-construction material (e.g. walls from bottles)



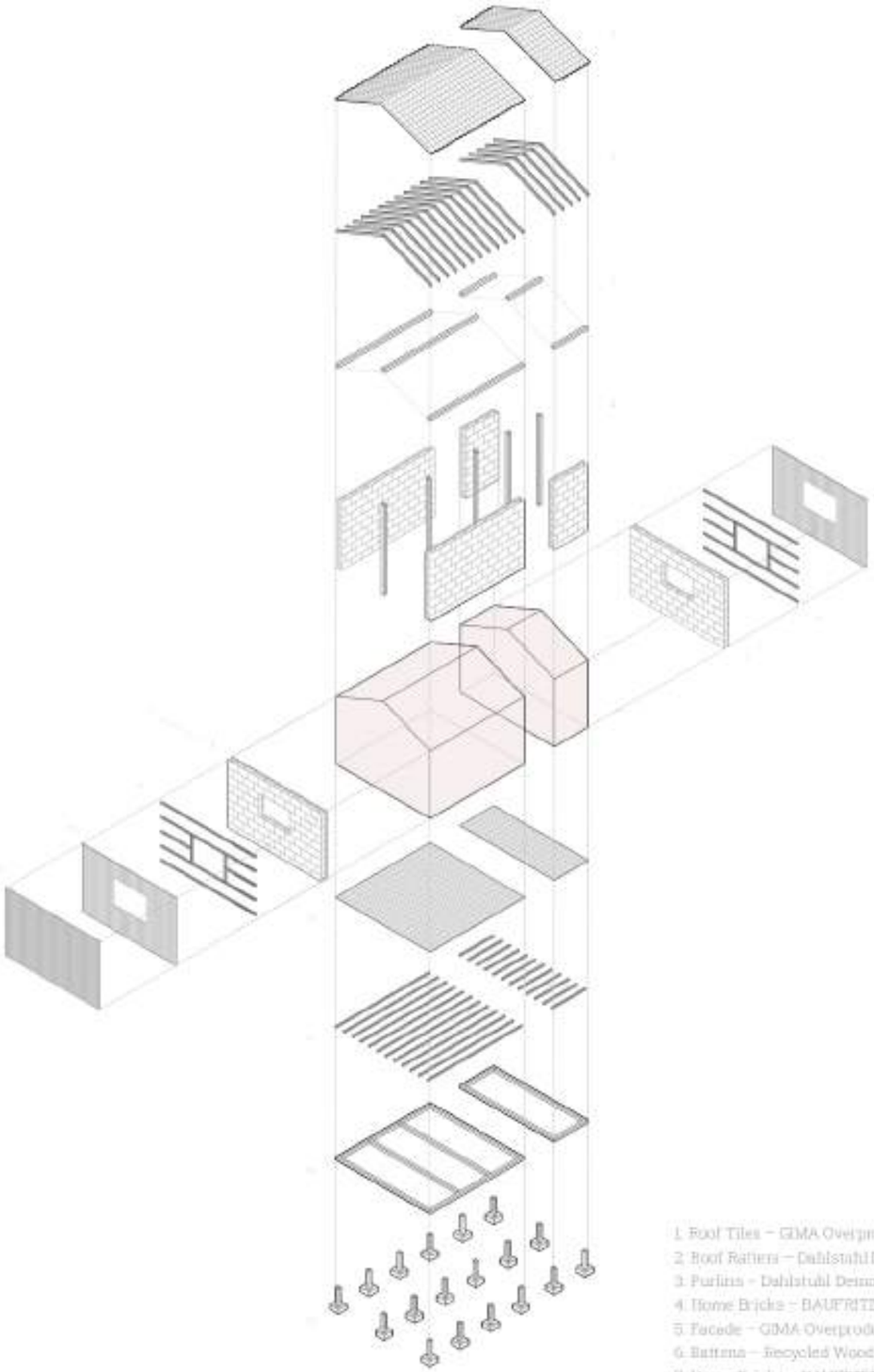
4 BUILDING MATERIAL MARKETPLACES

- Concular (certified building components from demolition projects)
- Ebay Marketplace (online marketplace with private sellers)
- Treibgut (Local on-site marketplace in Munich)

5 DEMOLITION PROJECTS

- Office demolition in Kaufbeuren by BBM Mahler demolition company
- Church demolition in Füssen by the municipal authority
- Roof framework demolition by a private individual

MATERIALS AVAILABLE IN THE CIRCULAR ECONOMY
– COLLECTED MATERIAL SAMPLES



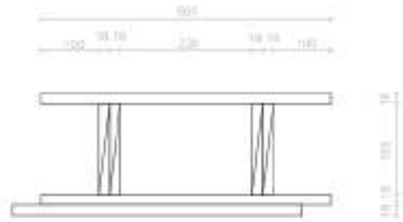
EXPLODED DIAGRAM
OF THE COMPONENTS 1:50

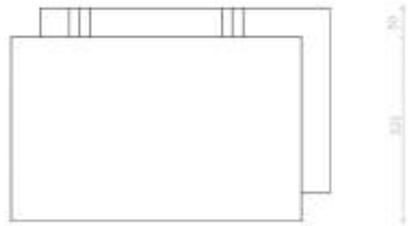
1. Roof Tiles – GIMA Overproduction
2. Roof Rafters – Dahlstuhl Demolition Lichtenhaag
3. Purlins – Dahlstuhl Demolition Lichtenhaag
4. Home Bricks – BAUFRITZ Offcuts
5. Facade – GIMA Overproduction
6. Battens – Recycled Wood Workshop Riedle + Bader
7. Home Bricks – BAUFRITZ Offcuts
8. Home Bricks – BAUFRITZ Offcuts
9. Battens – Recycled Wood Workshop Riedle + Bader
10. Wooden Shingle Facade – Church Demolition Füssen
11. Wooden Shingle Facade – Church Demolition Füssen
12. Mahogany Parquet – Church Demolition Füssen
13. Battens – Barracks Demolition
14. Timber Beams – Barracks Demolition
15. Oak Logs, Concrete Slabs – Cow Barn Demolition

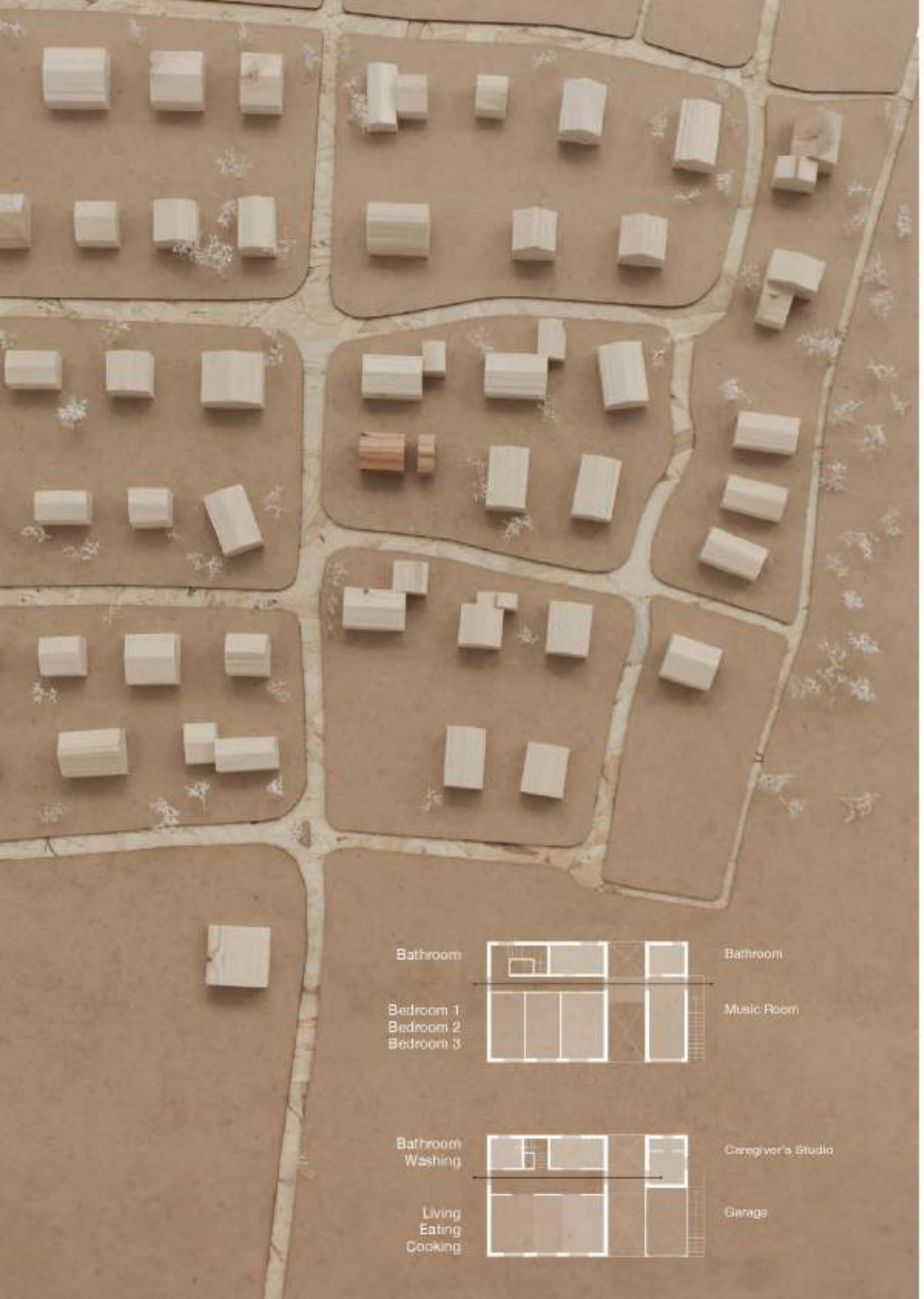
The „Home Bricks" are a modular building system for load-bearing walls. Their small-scale design makes it possible to incorporate leftover materials into the construction process. The modular structure allows the wooden bricks to be assembled directly on-site without specialized knowledge, simplifying the construction process and enabling the involvement of local helpers.

A key aspect of the "Home Bricks" system is the collaboration with the company Baufritz, which supports the project by providing wood offcuts from their prefabricated house production free of charge. Since June 2024, Baufritz collects, cuts, and delivers the offcuts in the required dimensions directly to the construction site.

The Home Bricks are made from low-emission Elka boards, produced using wood from sustainably managed regional forests. Their diffusion-open properties eliminate the need for additional vapor barriers. The exterior facade can be mounted directly onto the Home Bricks, while the interior surface allows flexible design options, such as clay plaster or a raw finish. With a patent secured by „LuxHome," the Home Bricks meet all requirements for fire safety, soundproofing, thermal insulation, and structural stability.

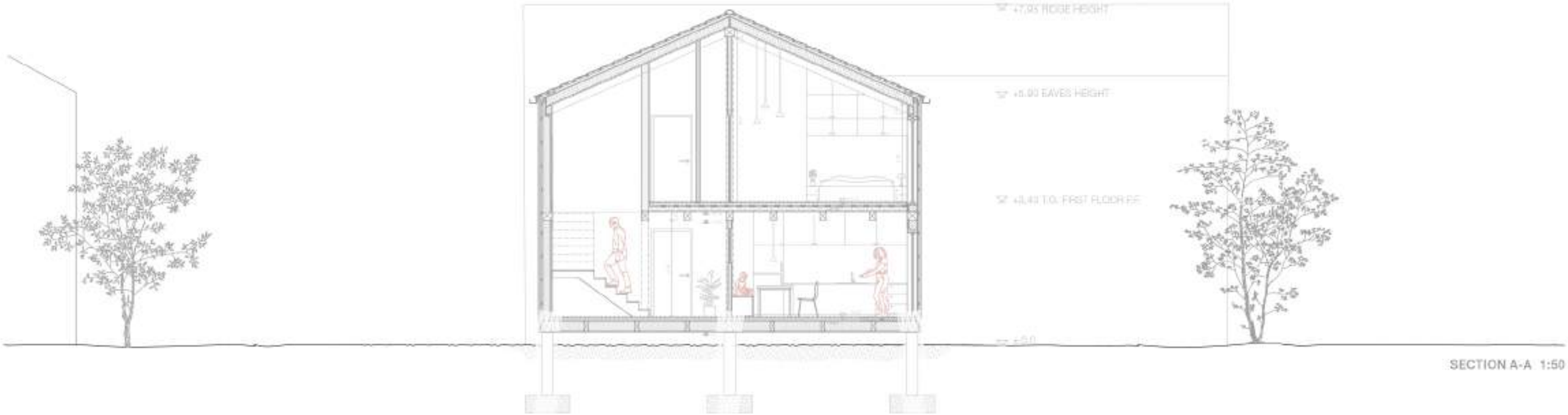




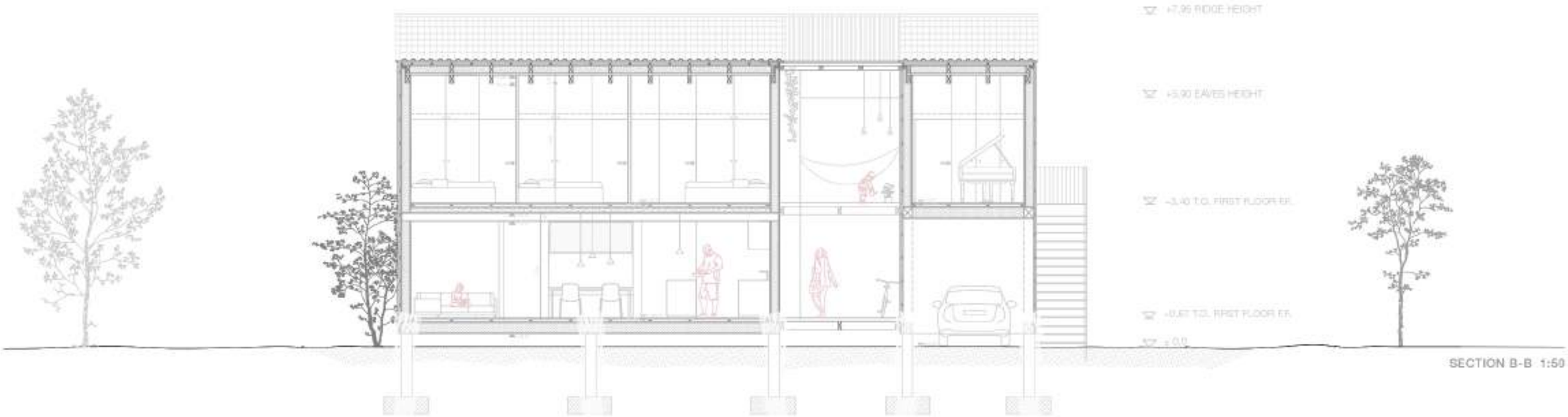


FIRST FLOOR PLAN 1:50

GROUND FLOOR PLAN 1:50



SECTION A-A 1:50



SECTION B-B 1:50

The mahogany flooring from the former church "Zu den Acht Seligkeiten" in Füssen demonstrates the potential of urban mining to recover high-quality building materials for integration into new architectural projects. Following the deconsecration of the church, the opportunity arose to salvage the valuable wood before demolition and repurpose it in new contexts, including the design of the „Recycle Residence".

In a collaborative event organized by Supertecture, over 300 m² of flooring were carefully dismantled by the team and 30 dedicated volunteers. The initiative was supported by Hilti, who provided tools for the task, and documented by an ARTE film crew for an upcoming documentary.

The recovered mahogany was transported to the reclaimed wood workshop of Riedle and Bader. There, it was processed for reuse through planing, sanding, and cutting as required. By incorporating this material into new construction projects, it is given a second life, supporting sustainable and resource-efficient building practices.

This event also marked the launch of Supertecture's „Seminarchitecture" initiative, a series of participatory workshops that invite volunteers to engage in urban mining projects. Through this program, Supertecture emphasizes a hands-on approach to fostering circular architecture and highlights the importance of reusing valuable building materials in contemporary design.

CREATING THE FRAMEWORK	FOUNDING OF THE CORPORATION	SUPERTECTURE 2 – FORMING PARTNERSHIPS AND PUBLIC RELATIONS		
	DESIGN PROCESS AT THE TUM	APPROVAL PLANNING	EXECUTION PLANNING	CONSTRUCTION PROCESS

Q1 Jan, Feb, Mar — **Q2** Apr, May, Jun — **Q3** Jul, Aug, Sep — **Q4** Oct, Nov, Dec — **Q5** Jan, Feb, Mar — **Q6** Apr

2024 — **2025**

- Client + project finding
- First kick-off meeting
- Start of collaboration at TUM with Kristina
- Decision to establish Supertecture 2
- Visiting LuxHome: Obtaining the Home Brick Patent
- 1st project presentation, approval of the Irsee municipal council
- Visiting GIMA, Concular, elka
- Closing of new partnerships
- Visiting wood workshop Riedle & Bader: Letter of intent signed
- First project presentation at the Baufritz Transformathon
- Submission of the Bachelor thesis at TUM
- BAUFRITZ now collects wood offcuts for us free of charge!
- Start of filming a documentary with arte
- Church demolition in Füssen: Urban mining of mahogany flooring
- Project presentation at the Climate Summit: 1st place in the BAUFRITZ Transformathon
- Building permit application is submitted
- Pavilion is built as temporary storage in Irsee
- Launch of the Supertecture lecture series on sustainable architecture
- Beginning of construction

INSIDE OUT

03

Uta Graff, +49 8928922499
Studio Schneider Türtscher,
7th Semester

MY PARTICIPATION IN THE PROJECT:

Conceptual design, physical modelmaking, site analysis, concept sketches, 1:100 floor plans and sections in ArchiCAD

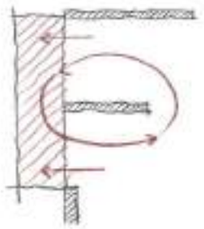
Pauline G., Luciana R., Eva Marie S.

As part of the semester project „Intermezzo," in collaboration with the guest professorship of Schneider-Türtscher from Zurich, the project investigates the traces of everyday life within existing buildings. Focusing on a medium-sized residential building from post-war Munich, the aim is to reframe its „rigid" architecture by offering new perspectives. The project explores how architectural interventions can evolve to meet current living standards and social needs while maintaining the identity and structural coherence of the original design.

Through subtle modifications to the existing layout, the design enhances the spatial quality of the building. By removing certain non-bearing walls, facades and corridors are opened up, creating new circulation routes that promote better movement and interaction. The balconies are reimagined with dynamic forms and supported by a light, self-supporting hybrid structure made of wood and steel. The design of the form adapts to the surrounding environment, offering residents a direct connection to nature. This approach transforms the building from a static structure into an open, flexible space that fosters a deeper engagement with its surroundings. The project features diverse apartment configurations, from efficient one-bedroom units to spacious family homes and co-living spaces for up to eight residents.

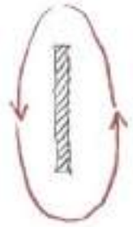
SITE MODEL 1:500 BY LUCIANA

The ground floor is transformed into a kindergarten, with the main entrance relocated from the east to the west facade. A sheltered inner courtyard provides outdoor space for both the kindergarten and the residents. Through careful topographical adjustments, the existing green spaces are given new meaning, while the narrow path leading to the backyard highlights the intimate nature of the environment. This creates a series of atmospheric spaces, new visual connections, and a protected outdoor area for the children.



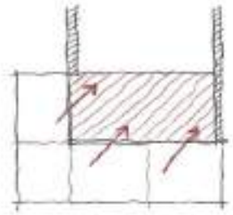
MERGING INTERIOR AND EXTERIOR

The addition of lightweight, self-supporting wooden balconies connects indoor living spaces to the surrounding nature, creating a seamless interaction between the interior and the environment.



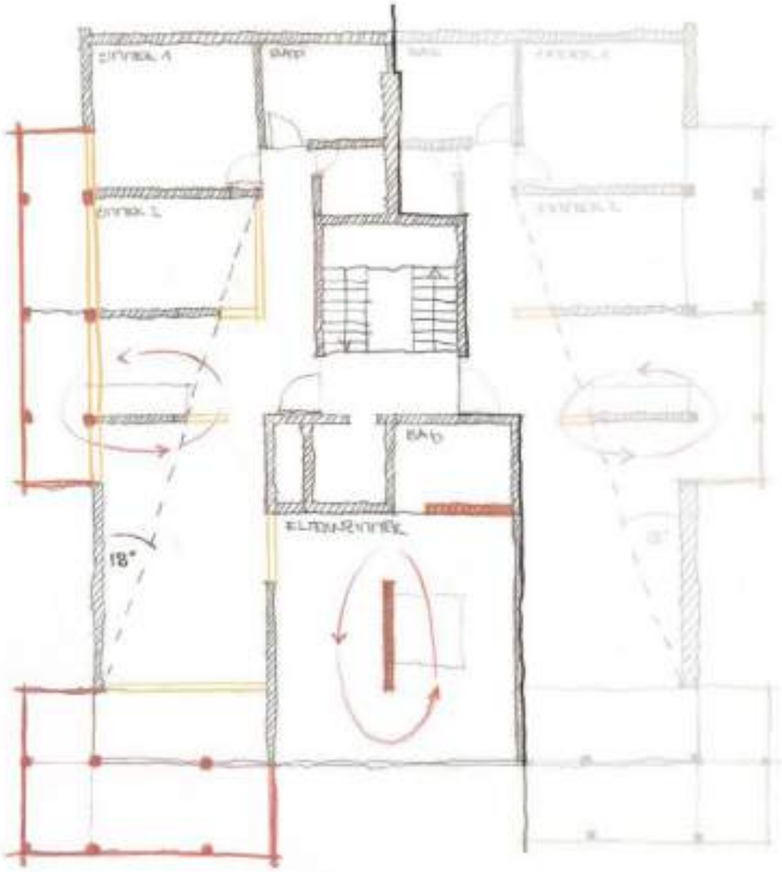
CREATING DYNAMIC CIRCULATION

By shortening selected walls in the existing layout, new circulation routes are created. These subtle changes enhance the building's spatial flow by improving movement patterns and increasing the functionality of the spaces.

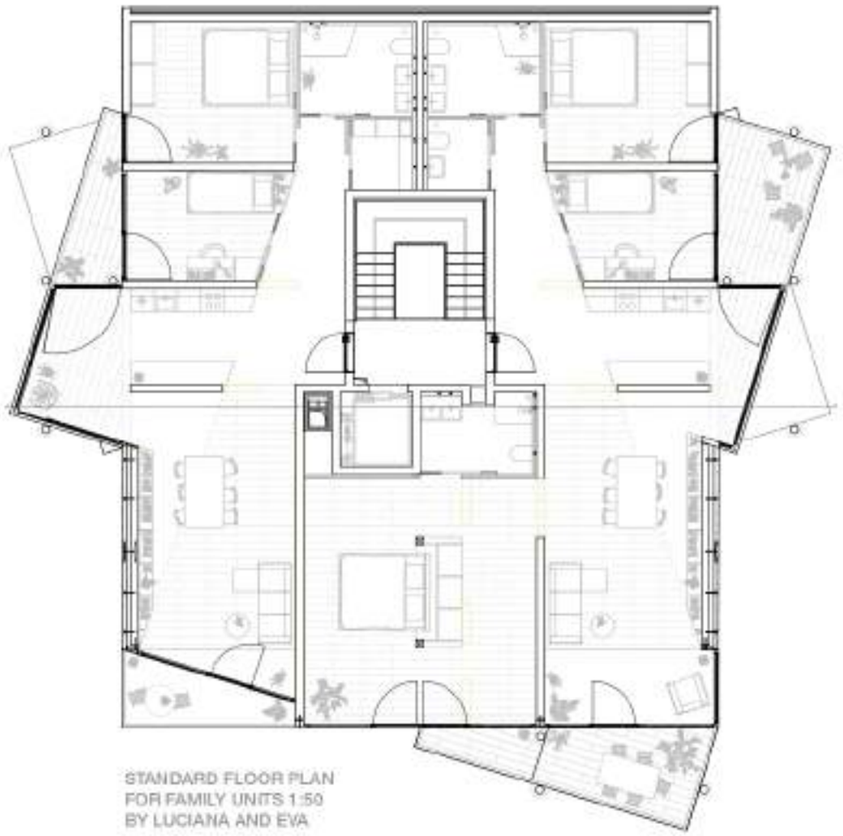


MAXIMIZING LIGHT AND FRAMING VIEWS

Selective structural adjustments enhance visual connections to the surrounding greenery. By inviting natural light into the interiors, the spatial quality is improved while enhancing the sense of openness.



MODEL OF A FAMILY APARTMENT 1: 25
BY LUCIANA AND PAULINE



STANDARD FLOOR PLAN
FOR FAMILY UNITS 1:50
BY LUCIANA AND EVA

FLOOR PLAN 1:100
SHARED FLAT FOR 8 PEOPLE

FLOOR PLAN 1:100
SINGLE STUDIO APARTMENTS

The design introduces a variety of apartment types, ranging fromcompact one-bedroomunitsto larger familyapartments and shared housing for up to eight people. Small yet impactful changes, such as shortening walls, optimize natural light and improve spatial usability. The integration of lightweight wooden balconies enhances the relationship between interior and exterior, fostering a continuous connection to nature while creating a harmonious flow throughout the building.

MODEL OF A FAMILY APARTMENT 1:25 – WEST FACADE

SOUTH FACADE VIEW 1:100 BY LUCIANA

SECTION A-A 1:100 BY EVA

MODEL OF A FAMILY APARTMENT
WEST FACADE 1: 25

URBAN WETLANDS

04

Prof. Stephen Bates, +49892892311
Studio Krucker Bates,
4th Semester

The semester project led by Studio Krucker Bates at the Technical University of Munich focuses on urban and landscape design in the suburban belt of Munich. At the heart of the design is the concept of an „Urban Wetland" – an ecosystem that promotes biodiversity, improves water quality, and regulates the microclimate in urban areas. The integration of this Urban Wetland represents a sustainable approach to enhancing both ecological and living quality in the urban environment.

The design includes a detailed exploration of the Urban Wetlands and the routing of pathways within the urban context. Straight, paved paths improve the connectivity to the surrounding university campus by linking public transport infrastructure and residential areas to the campus. In addition, dynamically curved wooden walkways create an intimate atmosphere for pedestrians. The walkways create moments where the wooden paths blend seamlessly with the natural surroundings, offering visitors an immersive experience of the wetland. Through topographical adjustments and a revised planting scheme, the contrasting qualities of the pathways are accentuated.

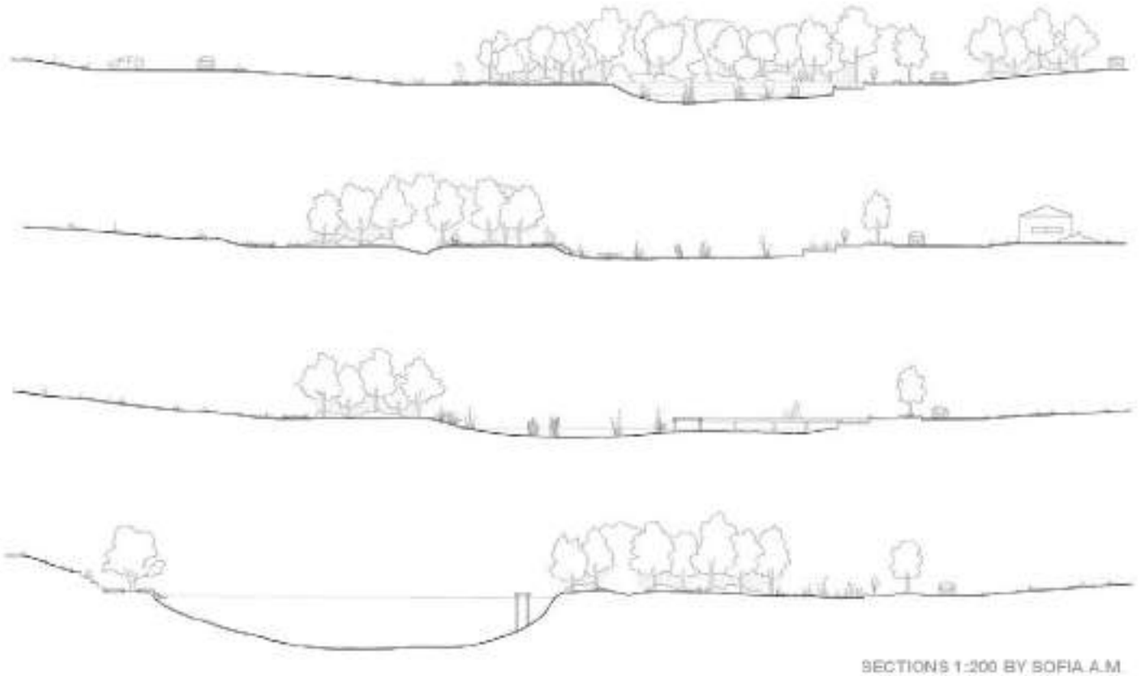
The bus stop is equipped with a shelter, bicycle racks are installed, and new seating areas for pedestrians are introduced. These interventions do not only improve the functionality but also enhance the overall quality of stay in this location.

MY PARTICIPATION IN THE PROJECT:

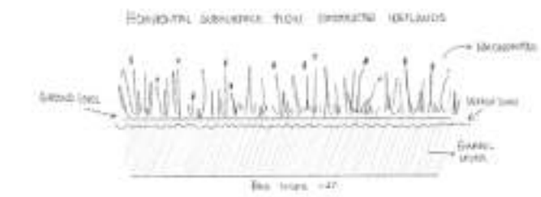
Conceptual design, physical modelmaking, site plan in Adobe Illustrator, concept sketches, wetland research, small moment floor plans and sections in ArchiCAD

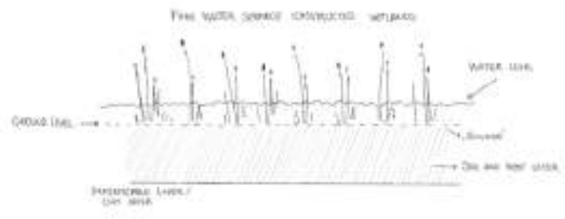
Sofia A.M., Pauline G., Luciana R.

SITE PLAN 1:200 BY LUCIANA R.

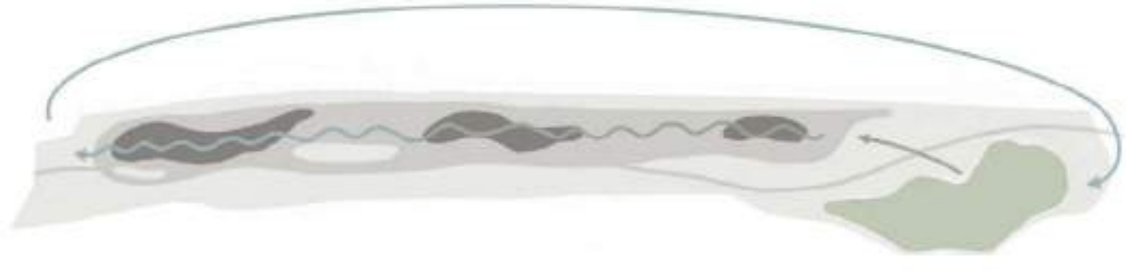


SECTIONS 1:200 BY SOFIA A.M





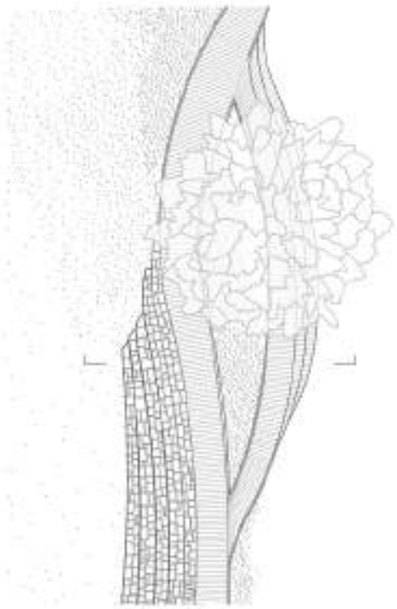
The design of the Urban Wetland offers a wide range of ecological benefits. It significantly contributes to the biodiversity of plants, animals, and microorganisms. At the same time, it serves as a natural water filter, reducing flood risks and improving the water quality. The wetland's ability to filter pollutants also enhances the air quality, while its evaporation and cooling processes help regulate the urban microclimate. In addition to these ecological functions, the wetland provides green spaces that offer atmospheric pathways for the residents of Freising.

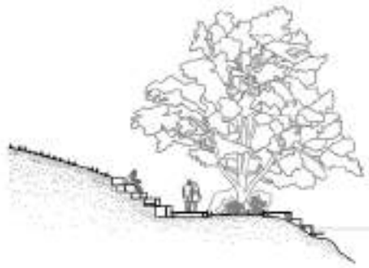


THE URBAN WETLAND CONCEPT: A SELF-SUSTAINING ECOSYSTEM. BY PAULINE G.

SMALL MOMENT MODEL 1:50 BY PAULINE G. AND LUCIANA R.

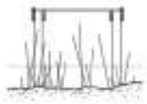
SMALL MOMENT VISUALIZATION
FLOOR PLAN 1:50
SECTION 1:50
BY LUCIANA R.

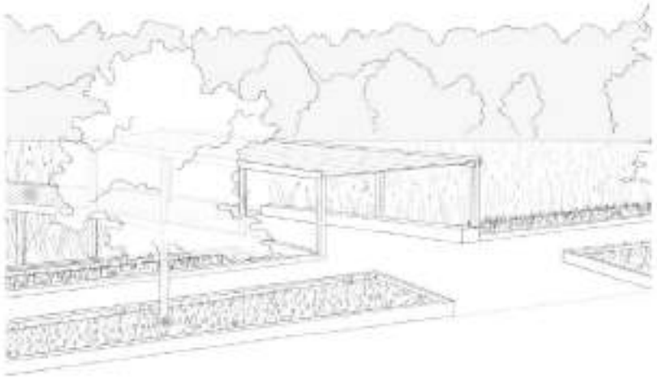


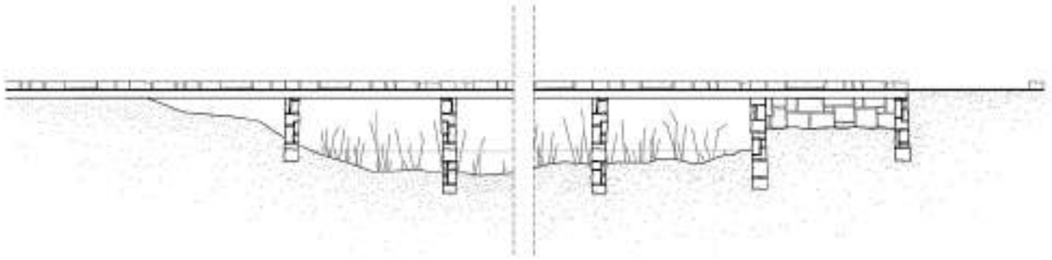


The „Small Moments" are designed to create spaces where pedestrians can pause and connect with the Urban Wetland. The natural qualities of the area are highlighted by the pathway design, offering opportunities for pause and reflection. The wooden walkways curve around the trees, while seating made from locally sourced stone is placed on previously unused slopes. This approach fully utilizes the potential of the natural environment, enhancing the overall experience of the space.

THE TWO TYPES OF PATHWAYS
SECTIONS 1:20 BY SOFIA A.M.







BUS STOP MODEL 1:20
BY PAULINE G. AND LUCIANA R.

THE TETRIS HOUSE

05

Prof. Florian Musso,
+49 8928922302
3rd Semester

The semester project „Tetris House", undertaken at the Chair of Building Construction and Materials under Prof. Florian Musso, explores innovative approaches to urban densification. The design features a Plus-Energy building that combines residential and commercial functions, achieving remarkable energy efficiency and sustainability.

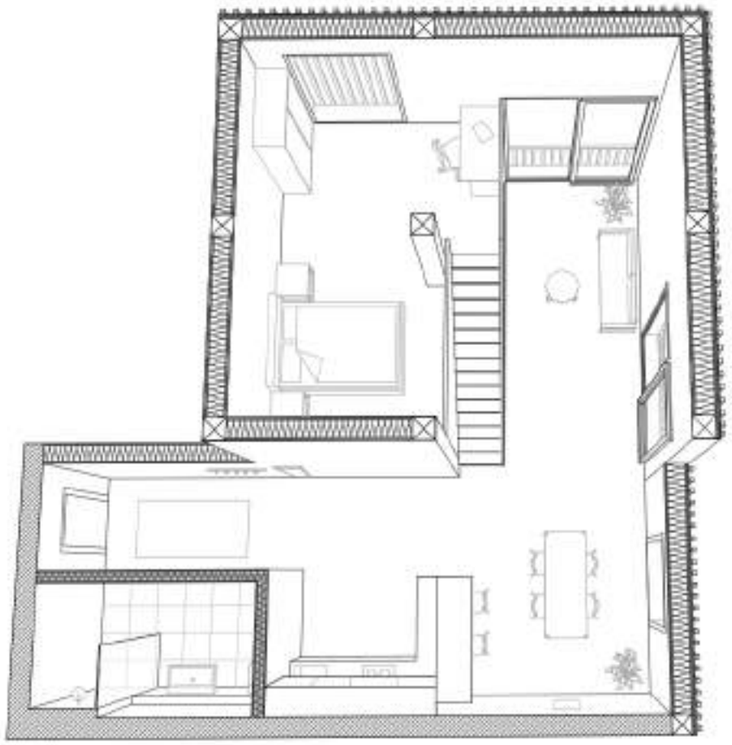
The ground floor is dedicated to a café, while the upper levels house diverse two-story apartments. These interlocking units maximize natural light and create spacious, functional living spaces, serving as a model for compact urban housing. The apartments are connected through a central reinforced concrete core, supported by a modular timber column structure, emphasizing both durability and material efficiency.

The two-story L-shaped layout is not only expressed in the building's floor plans, but also in the sections and in the facade. The exterior design incorporates treated wooden slats using the Yakisugi method, enhancing durability while providing shading in summer and heat retention in winter. The space beneath each staircase is optimized with built-in kitchen elements or storage cabinets, ensuring functionality while contributing to a comfortable and cohesive living environment.

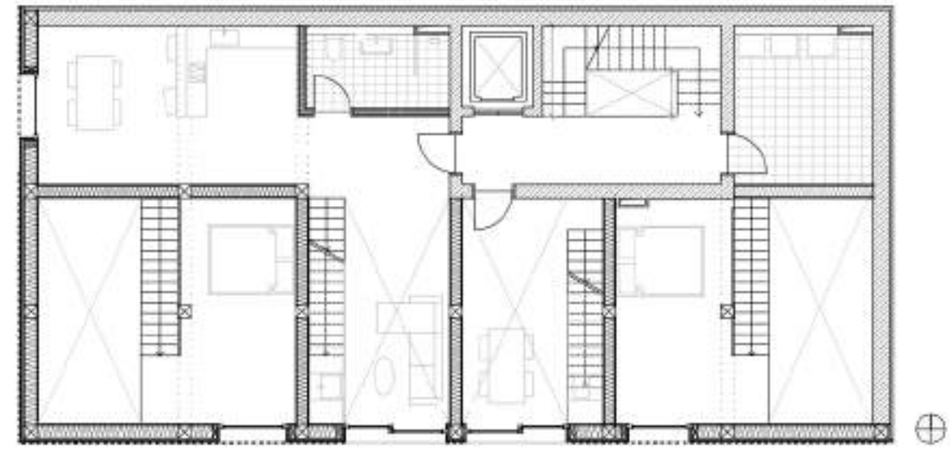
MY PARTICIPATION IN THE PROJECT:

Conceptual design, physical modelmaking, three panel projections, floor plans, ArchiCAD 3D render model

Pauline G., Luciana R., Mara S.

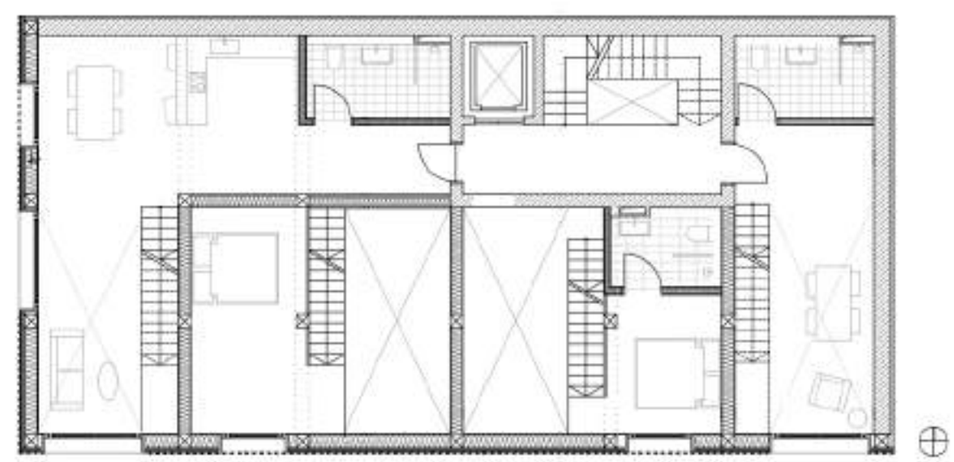


THREE DIMENSIONAL FLOOR PLAN 1:100 BY LUCIANA R.

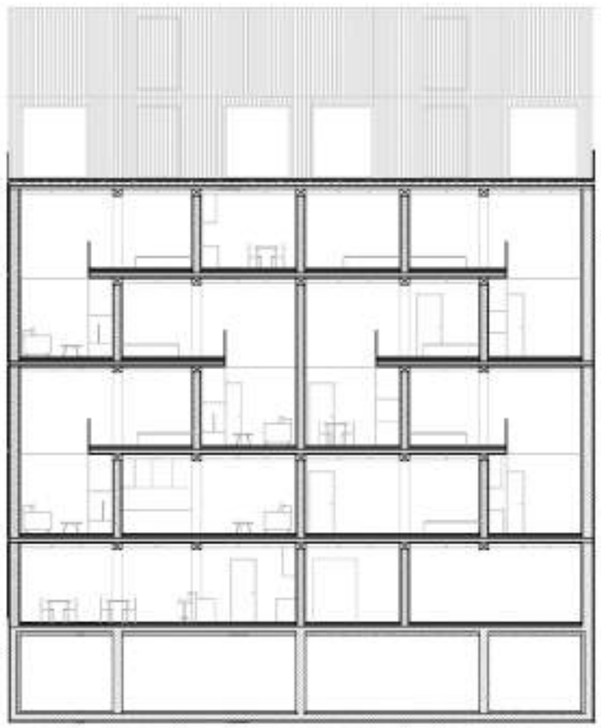


SECOND FLOOR PLAN 1:100 BY MARA S. AND LUCIANA R.

The floor plans follow clear principles of two-story design, forming an L-shaped configuration. This concept is reflected not only in the floor plans but also in the sectional design and façade. Space beneath each staircase is optimized with integrated kitchen elements or living room cabinets, ensuring efficient use of space while fostering a comfortable and inviting living atmosphere.

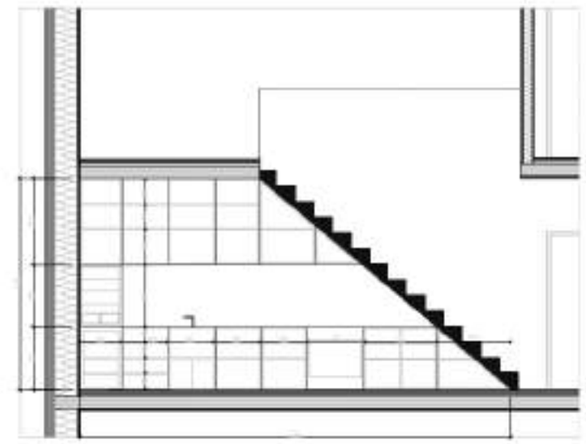


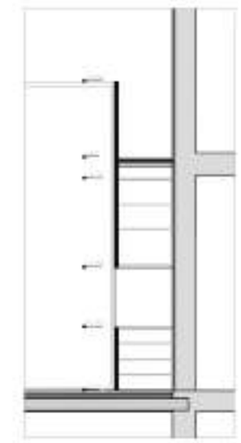
FIRST FLOOR PLAN 1:100 BY MARA S. AND LUCIANA R.

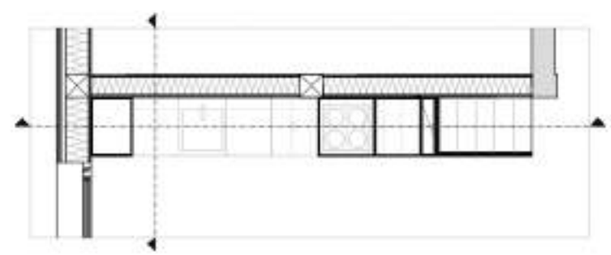


1:100 MODEL
SECTION A-A 1:100
BY PAULINE G. AND LUCIANA R.

The facade is composed of wooden lamellas treated with the Yakisugi method, enhancing the durability and weather resistance of the material. This traditional technique not only protects the wood but also contributes to the building's thermal efficiency. In summer, the lamellas provide effective shading, while in winter, their dark surface enhances heat absorption, improving energy efficiency. Openings in the facade reveal the building's two-story structure, highlighting the layered organization of the interiors.







THREE-PANEL PROJECTION:
THE BULTHAUP B3 KITCHEN
BY LUCIANA R. AND MARA S.